PADHAKU
Mathematics · Class 10 Mathematics

CHAPTER 1

Real Numbers

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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In-text questions · 1.2 The Fundamental Theorem of Arithmetic

? **Suppose your collection of primes includes all the possible primes. Does it contain only a finite number of integers, or infinitely many?**

✔ ANSWER

There are infinitely many primes. Multiplying them in all possible ways (with repeats allowed) gives an infinite collection of numbers: all the primes and all products of primes.

Infinitely many.

? **Can we produce all the composite numbers this way? Do you think that there may be a composite number which is not the product of powers of primes?**

✔ ANSWER

Yes, every composite number can be produced this way.

Factorising numbers (for example $32760 = 2^3 \times 3^2 \times 5 \times 7 \times 13$, or $123456789 = 3^2 \times 3803 \times 3607$) always gives a product of powers of primes.

No, there is no composite number that is not a product of powers of primes. This is the Fundamental Theorem of Arithmetic: every composite number can be written as a product of primes, uniquely apart from the order of the factors.

📝 Exercise 1.1

1 (i) **Express 140 as a product of its prime factors.**

✔ ANSWER

$140 = 2 \times 70 = 2 \times 2 \times 35 = 2 \times 2 \times 5 \times 7$

$140 = 2^2 \times 5 \times 7$

1 (ii) **Express 156 as a product of its prime factors.**

✔ ANSWER

$156 = 2 \times 78 = 2 \times 2 \times 39 = 2 \times 2 \times 3 \times 13$

$156 = 2^2 \times 3 \times 13$

1 (iii) **Express 3825 as a product of its prime factors.**

✔ ANSWER

$3825 = 3 \times 1275 = 3 \times 3 \times 425 = 3 \times 3 \times 5 \times 85$
$= 3 \times 3 \times 5 \times 5 \times 17$

3825 = $3^2 \times 5^2 \times 17$

1 (iv) **Express 5005 as a product of its prime factors.**

✔ ANSWER

5005 = 5 × 1001 = 5 × 7 × 143 = 5 × 7 × 11 × 13

5005 = 5 × 7 × 11 × 13

1 (v) **Express 7429 as a product of its prime factors.**

✔ ANSWER

7429 = 17 × 437 = 17 × 19 × 23

7429 = 17 × 19 × 23

2 (i) **Find the LCM and HCF of 26 and 91 and verify that LCM × HCF = product of the two numbers.**

✔ ANSWER

26 = 2 × 13, 91 = 7 × 13

HCF = product of the smallest powers of common primes = 13

LCM = product of the greatest powers of all primes = 2 × 7 × 13 = 182

Verification: LCM × HCF = 182 × 13 = 2366; 26 × 91 = 2366. Equal, verified.

HCF = 13, LCM = 182

2 (ii) **Find the LCM and HCF of 510 and 92 and verify that LCM × HCF = product of the two numbers.**

✔ ANSWER

$510 = 2 \times 3 \times 5 \times 17$, $92 = 2^2 \times 23$

HCF = 2

LCM = $2^2 \times 3 \times 5 \times 17 \times 23 = 23460$

Verification: LCM × HCF = 23460 × 2 = 46920; 510 × 92 = 46920. Equal, verified.

HCF = 2, LCM = 23460

2 (iii) **Find the LCM and HCF of 336 and 54 and verify that LCM × HCF = product of the two numbers.**

✔ ANSWER

$336 = 2^4 \times 3 \times 7$, $54 = 2 \times 3^3$

HCF = $2 \times 3 = 6$

LCM = $2^4 \times 3^3 \times 7 = 16 \times 27 \times 7 = 3024$

Verification: LCM × HCF = 3024 × 6 = 18144; 336 × 54 = 18144. Equal, verified.

HCF = 6, LCM = 3024

3 (i) **Find the LCM and HCF of 12, 15 and 21 by the prime factorisation method.**

✔ ANSWER

$12 = 2^2 \times 3$, $15 = 3 \times 5$, $21 = 3 \times 7$

HCF = 3 (only 3 is common to all three)

LCM = $2^2 \times 3 \times 5 \times 7 = 420$

HCF = 3, LCM = 420

3 (ii) **Find the LCM and HCF of 17, 23 and 29 by the prime factorisation method.**

✔ ANSWER

17, 23 and 29 are all prime numbers: 17 = 17, 23 = 23, 29 = 29

No common prime factor, so HCF = 1

LCM = 17 × 23 × 29 = 11339

HCF = 1, LCM = 11339

3 (iii) **Find the LCM and HCF of 8, 9 and 25 by the prime factorisation method.**

✔ ANSWER

$8 = 2^3$, $9 = 3^2$, $25 = 5^2$

No common prime factor, so HCF = 1

LCM = $2^3 \times 3^2 \times 5^2 = 8 \times 9 \times 25 = 1800$

HCF = 1, LCM = 1800

4 **Given that HCF (306, 657) = 9, find LCM (306, 657).**

✔ ANSWER

HCF × LCM = product of the two numbers

LCM (306, 657) = (306 × 657) ÷ HCF = 201042 ÷ 9 = 22338

LCM (306, 657) = 22338

5 **Check whether 6^n can end with the digit 0 for any natural number n.**

✔ ANSWER

A number ends with the digit 0 only if it is divisible by 10 = 2 × 5, so its prime factorisation must contain the prime 5.

$6^n = (2 \times 3)^n = 2^n \times 3^n$. The only primes in its factorisation are 2 and 3.

By the uniqueness of the Fundamental Theorem of Arithmetic, no other prime (like 5) can appear in the factorisation of 6^n.

So 6^n can never end with the digit 0 for any natural number n.

6 **Explain why 7 × 11 × 13 + 13 and 7 × 6 × 5 × 4 × 3 × 2 × 1 + 5 are composite numbers.**

✔ ANSWER

A composite number has a factor other than 1 and itself.

7 × 11 × 13 + 13 = 13 × (7 × 11 + 1) = 13 × 78 = 1014. It has 13 as a factor, so it is **composite**.

7 × 6 × 5 × 4 × 3 × 2 × 1 + 5 = 5 × (7 × 6 × 4 × 3 × 2 × 1 + 1) = 5 × 1009 = 5045. It has 5 as a factor, so it is **composite**.

7 **Sonia takes 18 minutes to drive one round of a circular path, while Ravi takes 12 minutes. They start at the same point, at the same time, in the same direction. After how many minutes will they meet again at the starting point?**

✔ ANSWER

They meet again at the starting point after a time that is a multiple of both 18 and 12, i.e. after LCM (18, 12) minutes.

$18 = 2 \times 3^2$, $12 = 2^2 \times 3$

$LCM = 2^2 \times 3^2 = 36$

They will meet again at the starting point after 36 minutes.

📝 Exercise 1.2

1 **Prove that √5 is irrational.**

✔ ANSWER

1. Let us assume, to the contrary, that √5 is rational. Then $\sqrt{5} = a/b$, where a and b are coprime integers and $b \neq 0$.

2. So $b\sqrt{5} = a$. Squaring both sides: $5b^2 = a^2$.

3. Therefore 5 divides a^2. By Theorem 1.2 (p prime, p divides $a^2 \Rightarrow$ p divides a), 5 divides a.

4. So $a = 5c$ for some integer c. Substituting: $5b^2 = 25c^2$, i.e. $b^2 = 5c^2$.

5. So 5 divides b^2, and by Theorem 1.2, 5 divides b.

6. Thus a and b have 5 as a common factor. This contradicts the fact that a and b are coprime.

7. The contradiction arose from our wrong assumption that √5 is rational.

Hence √5 is irrational.

2 **Prove that 3 + 2√5 is irrational.**

✔ ANSWER

1. Let us assume, to the contrary, that 3 + 2√5 is rational. Then 3 + 2√5 = a/b, where a and b are coprime integers and b ≠ 0.

2. Rearranging: 2√5 = a/b − 3 = (a − 3b)/b, so √5 = (a − 3b)/2b.

3. Since a and b are integers, (a − 3b)/2b is rational. So √5 is rational.

4. But this contradicts the fact that √5 is irrational (Exercise 1.2, Q1).

5. So our assumption is wrong.

Hence 3 + 2√5 is irrational.

3 (i) **Prove that 1/√2 is irrational.**

✔ ANSWER

1. Assume, to the contrary, that 1/√2 is rational. Then 1/√2 = a/b, with a, b coprime integers, b ≠ 0 (and a ≠ 0).

2. Then √2 = b/a. Since a and b are integers, b/a is rational, so √2 is rational.

3. This contradicts the fact that √2 is irrational (Theorem 1.3).

4. So our assumption is wrong.

Hence 1/√2 is irrational.

3 (ii) **Prove that 7√5 is irrational.**

✔ ANSWER

1. Assume, to the contrary, that 7√5 is rational. Then 7√5 = a/b, with a, b coprime integers, b ≠ 0.

2. Then √5 = a/7b. Since 7, a and b are integers, a/7b is rational, so √5 is rational.

3. This contradicts the fact that √5 is irrational.

4. So our assumption is wrong.

Hence 7√5 is irrational.

3 (iii) **Prove that 6 + √2 is irrational.**

✔ ANSWER

1. Assume, to the contrary, that 6 + √2 is rational. Then 6 + √2 = a/b, with a, b coprime integers, b ≠ 0.

2. Then √2 = a/b − 6 = (a − 6b)/b. Since a and b are integers, (a − 6b)/b is rational, so √2 is rational.

3. This contradicts the fact that √2 is irrational (Theorem 1.3).

4. So our assumption is wrong.

Hence 6 + √2 is irrational.

🎉 **Chapter 1 done! Revise the answers once more tomorrow.**